Data Communication
CSE 2211

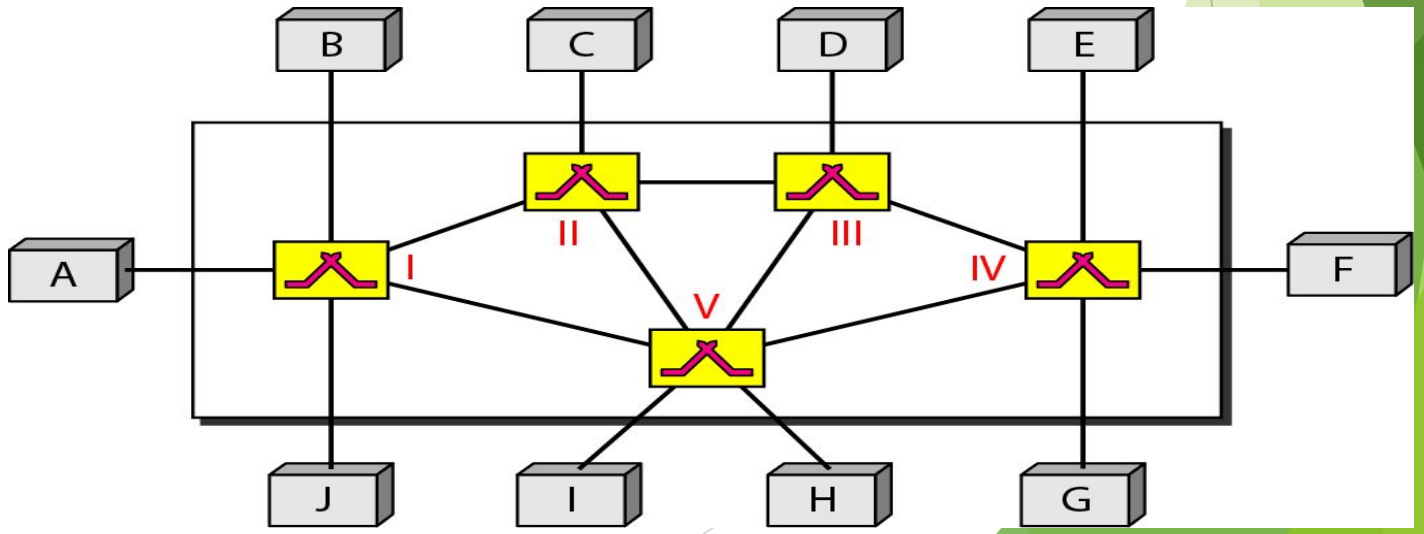
Chapter 8: Switching

Why Switching?

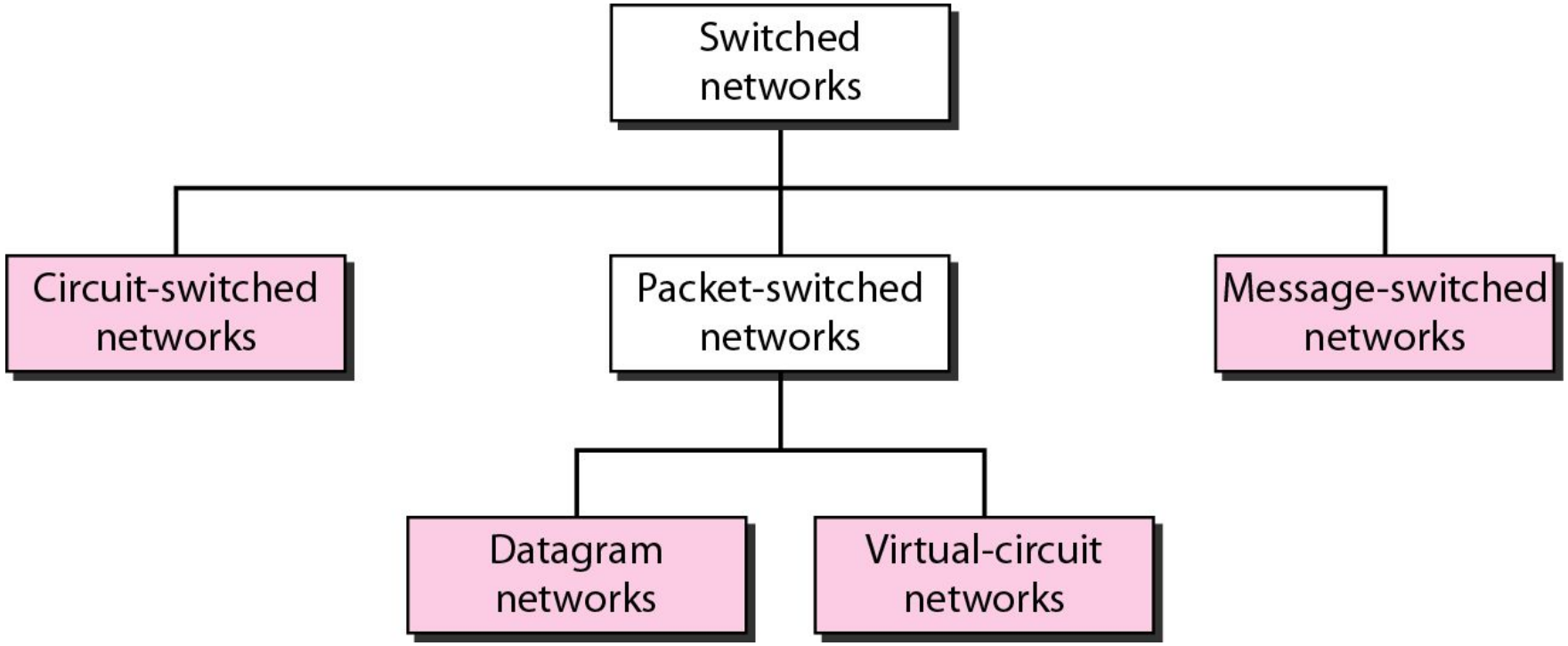
- A network is a set of connected devices. Some of the conventional ways of interconnecting devices are:
  a) Point-to point connection between devices as in a mesh topology.
  b) Connection between a central device and every other device as in star topology.
  c) Bus topology – not practical if the devices are at great distances.
- All these techniques require extensive cabling, dependence on a central server or a central bus.
  - impractical and wasteful when applied to very large networks
- The solution to this interconnectivity problem is switching.

Switching Network:

- A switched network consists of a series of interlinked nodes, called switches.
- Switches are devices capable of creating temporary connections between two or more devices linked to the switch.
- In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.

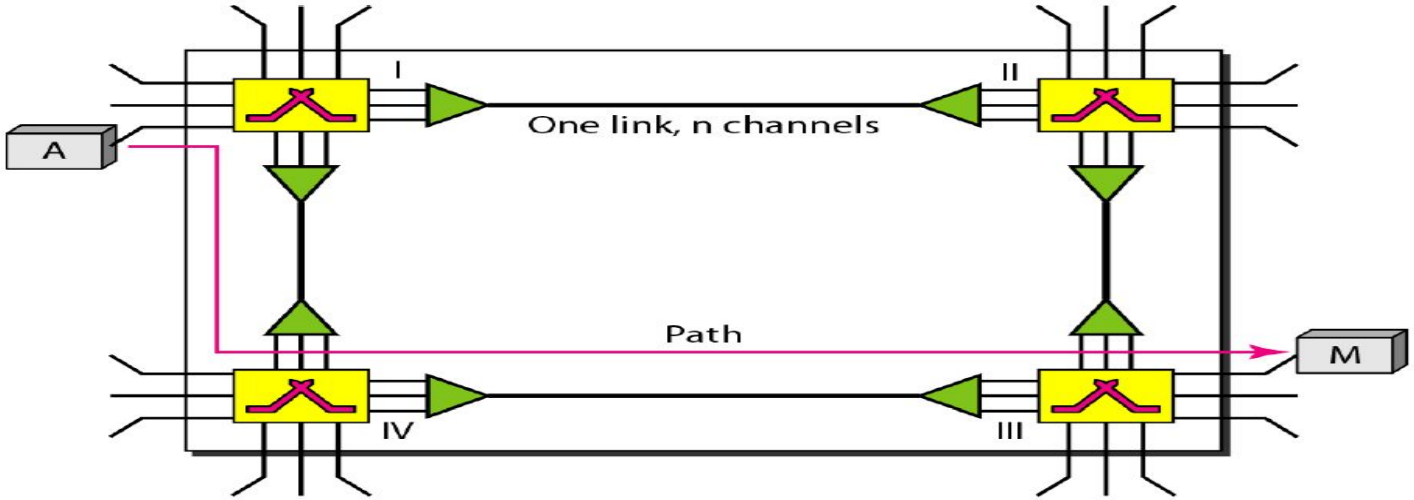


Types of Switching Network:



Circuit Switching

- **Circuit-switched network** consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links.
- Each connection uses only one dedicated channel on each link.
- Each link is normally divided into *n* channels by using FDM or TDM.



Key points of Circuit Switching

- Circuit switching takes place at the physical layer
- Before starting communication, make a reservation for the resources (Setup Phase)
  - resources, such as channels (bandwidth in FDM and time slots in TDM), switch buffers, switch processing time, switch I/O ports
- Data transferred between the two stations are not packetized (physical layer transfer of the signal).
- There is no addressing involved during data transfer. The switches route the data based on their occupied band (FDM) or time slot (TDM).
- There is end-to-end addressing used during the setup phase

Three phases of Circuit Switching

- ***Setup Phase***
  - Before the two parties can communicate, a dedicated circuit needs to be established.
  - connection setup means creating dedicated channels between the switches.
  - end-to-end addressing is required for creating a connection between two end systems. These can be, for example, the addresses of the computers assigned by the administrator in a TDM network, or telephone numbers in an FDM network.
- ***Data Transfer Phase***
  - After the establishment of the dedicated circuit (channels), the two parties can transfer data
- ***Teardown Phase***
  - When one of the parties needs to disconnect, a signal is sent to each switch to release the resources

Figure *Circuit-switched network used in Example*

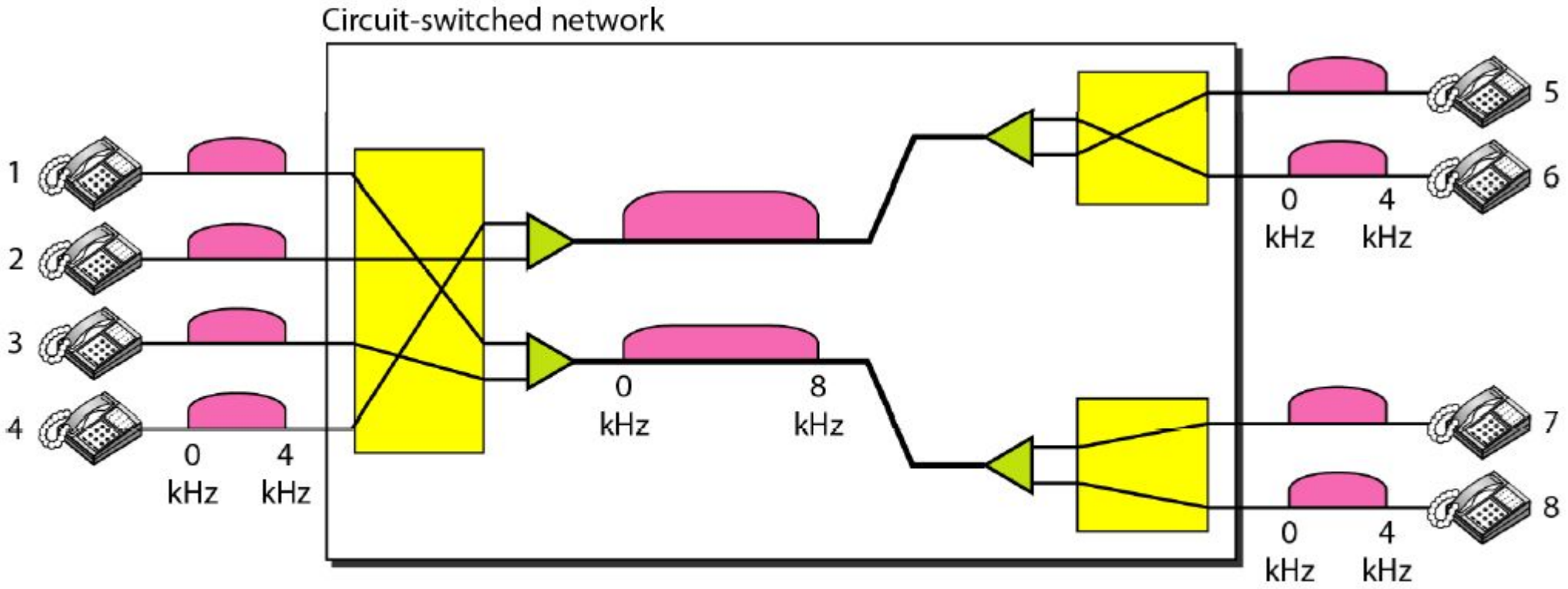
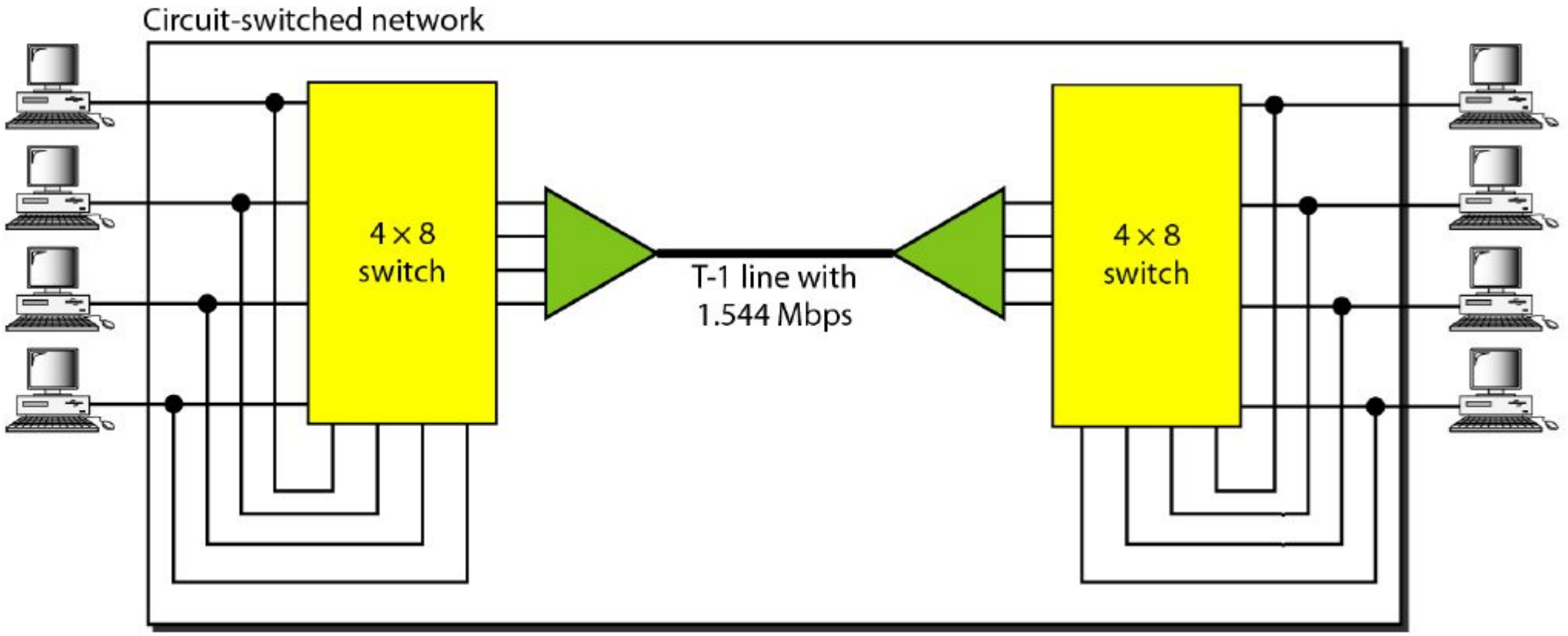


Figure 8.5 *Circuit-switched network used in Example 8.2*

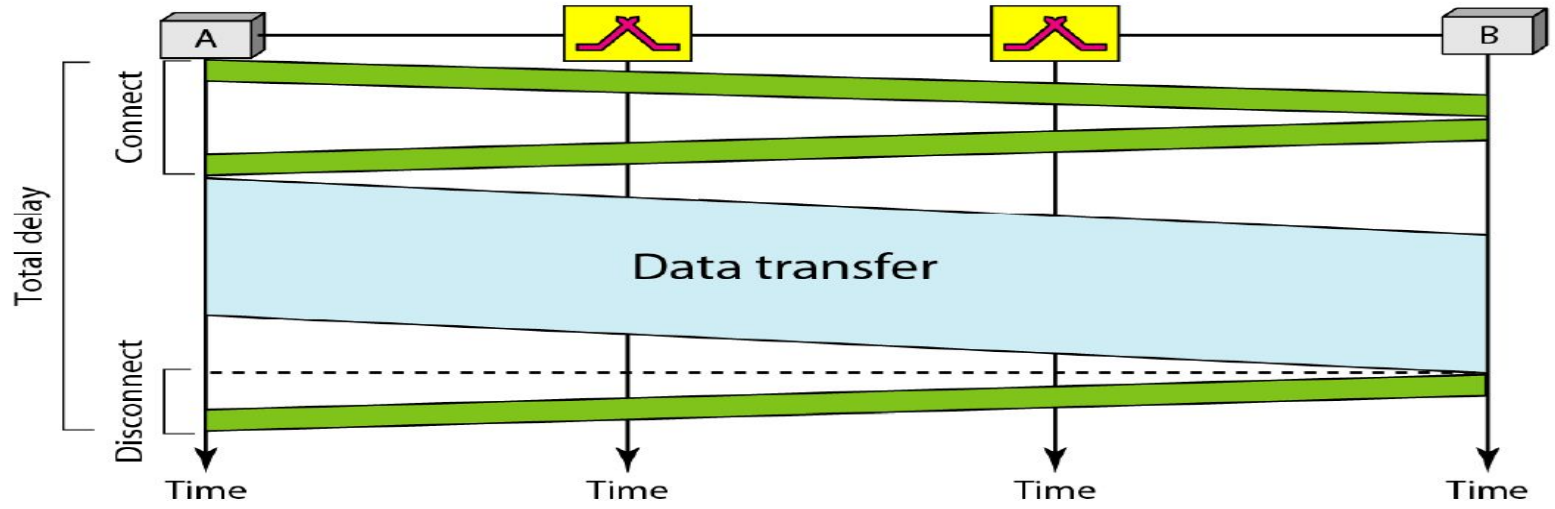


Efficiency of Circuit Switching

- Efficiency are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection and are unavailable to other connections.

Delay of Circuit Switching

- **Delay caused by the setup is the sum of four parts:**
  - the propagation time of the source computer request (slope of the first gray box),
  - the request signal transfer time (height of the first gray box),
  - the propagation time of the acknowledgment from the destination computer (slope of the second gray box),
  - the signal transfer time of the acknowledgment (height of the second gray box).
- **The delay due to data transfer is the sum of two parts:**
  - the propagation time (slope of the colored box)
  - data transfer time (height of the colored box), which can be very long.
- **The delay due to the time needed to tear down the circuit.**



Message Switching

- With message switching there is no need to establish a dedicated path between two stations.
- When a station sends a message, the destination address is appended to the message.
- Each node receives the entire message, stores it in its buffer, and then transmits the message to the next node.
- A message will only be delivered if the next hop and the link connecting it are both available, otherwise it'll be stored indefinitely.
- This type of network is called a store-and-forward network.

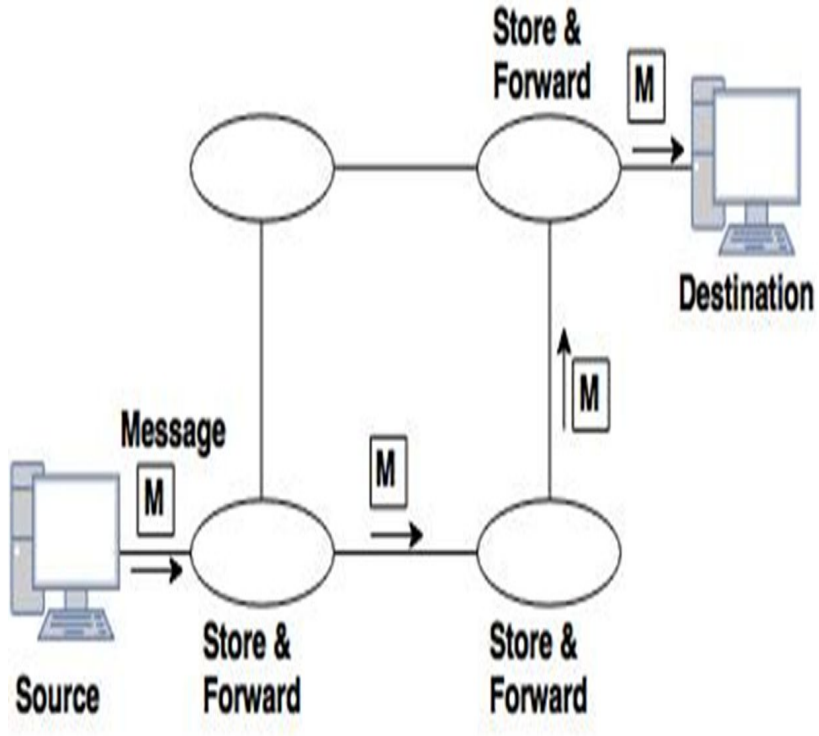


Fig: Message Switching

Packet Switching

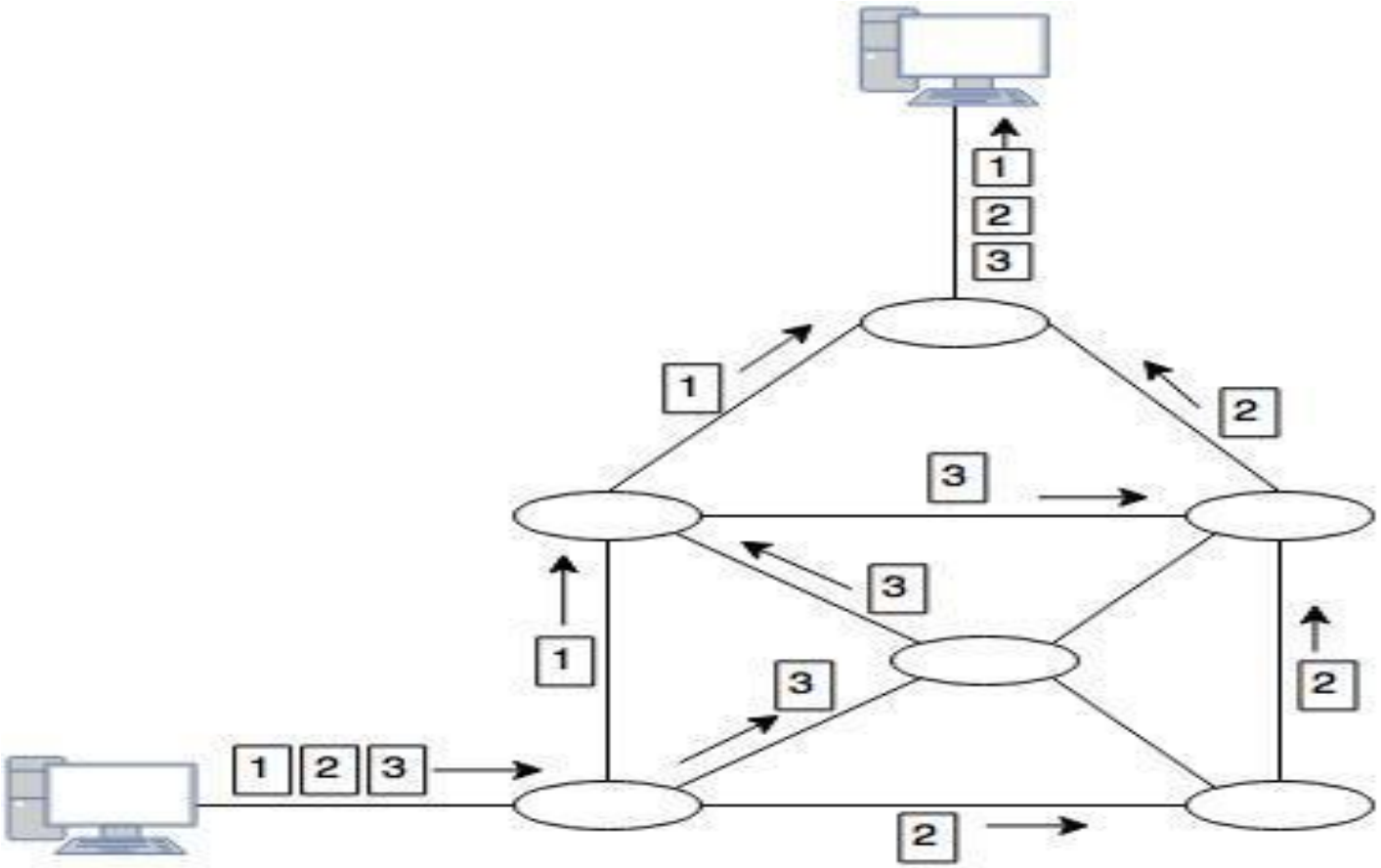


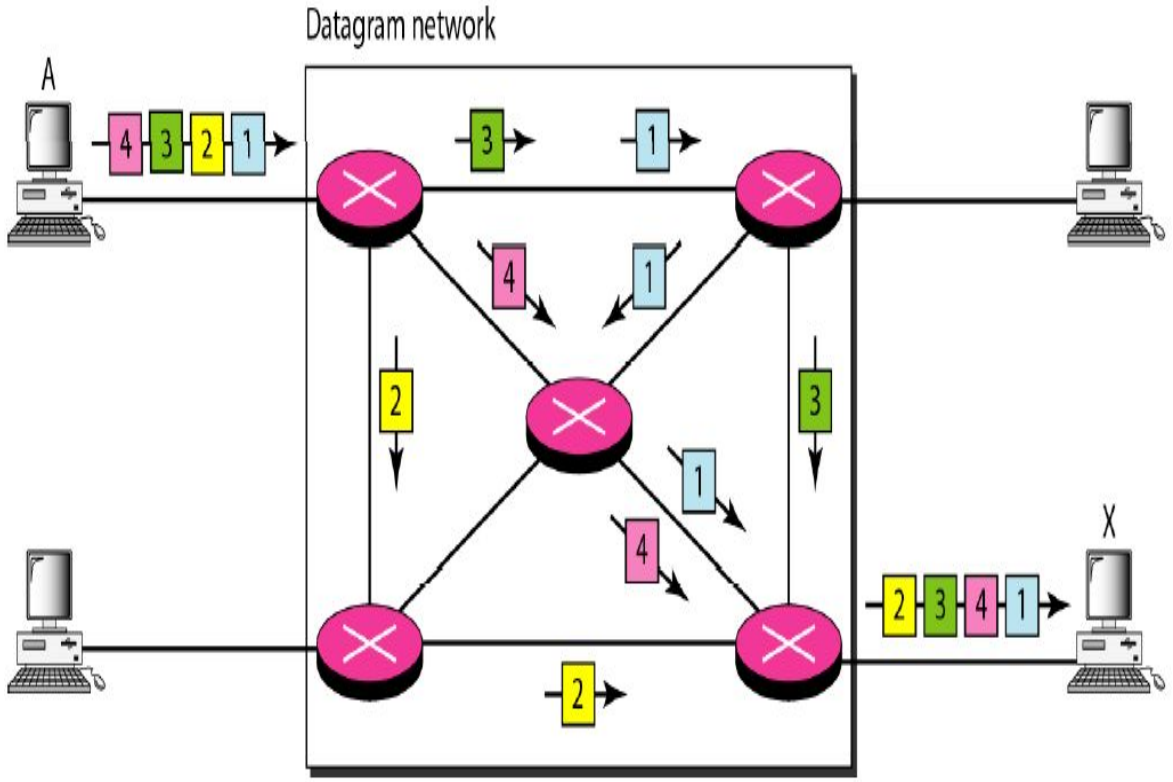
Fig: Datagram Packet Switching

Packet Switching

- In packet switching network, a message is broken into small parts, called packets.
- Each packet is tagged with appropriate source and destination addresses.
- There is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet. Resources are allocated on demand
- Packets are generally accepted onto the network on a first-come, first-served basis. If the network becomes overloaded, packets are delayed or discarded (``dropped‘’).
- There are two methods of packet switching: Datagram and virtual circuit.

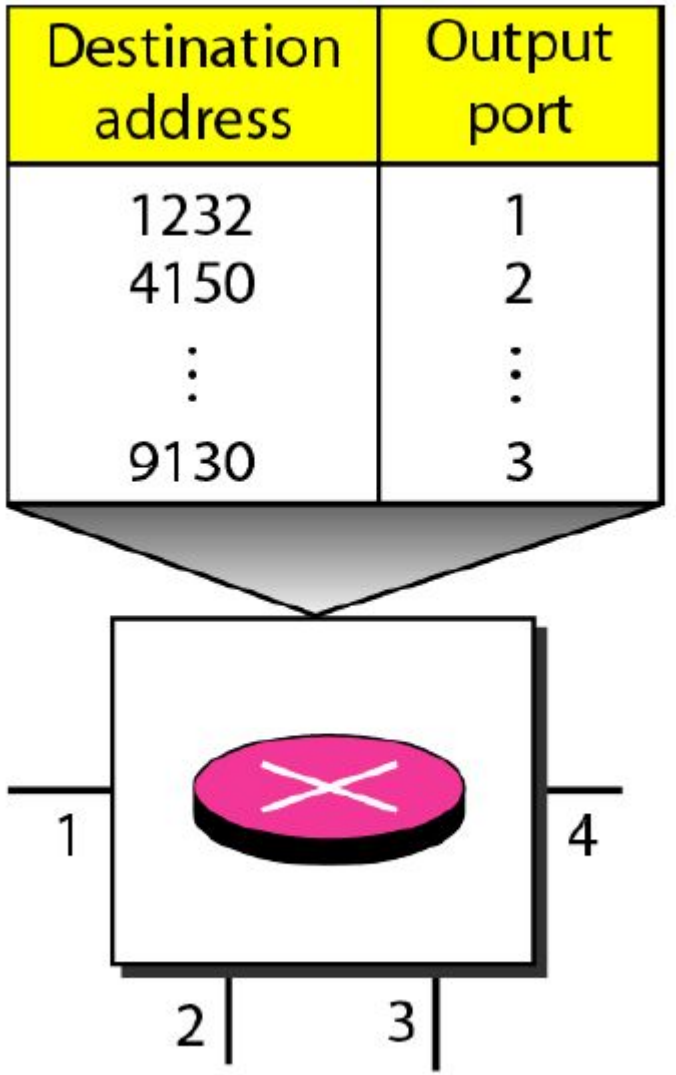
Datagram Networks

- In a **datagram network**, each packet is treated independently of all others.
- Even if a packet is part of a multipacket transmission, the network treats it as though it existed alone.
- Packets in this approach are referred to as datagrams.
- Datagram switching is normally done at the network layer.
- The switches in a datagram network are traditionally referred to as routers.
- The datagram networks are sometimes referred to as **connectionless networks**.



Routing Table

- If there are no setup or teardown phases, how are the packets routed to their destinations in a datagram network?
- Each switch has a routing table which is based on the destination address.
- The destination addresses and the corresponding forwarding output ports are recorded in the tables
- Every packet in a datagram network carries destination address of the packet.
- The routing table is consulted to find the corresponding port through which the packet should be forwarded.

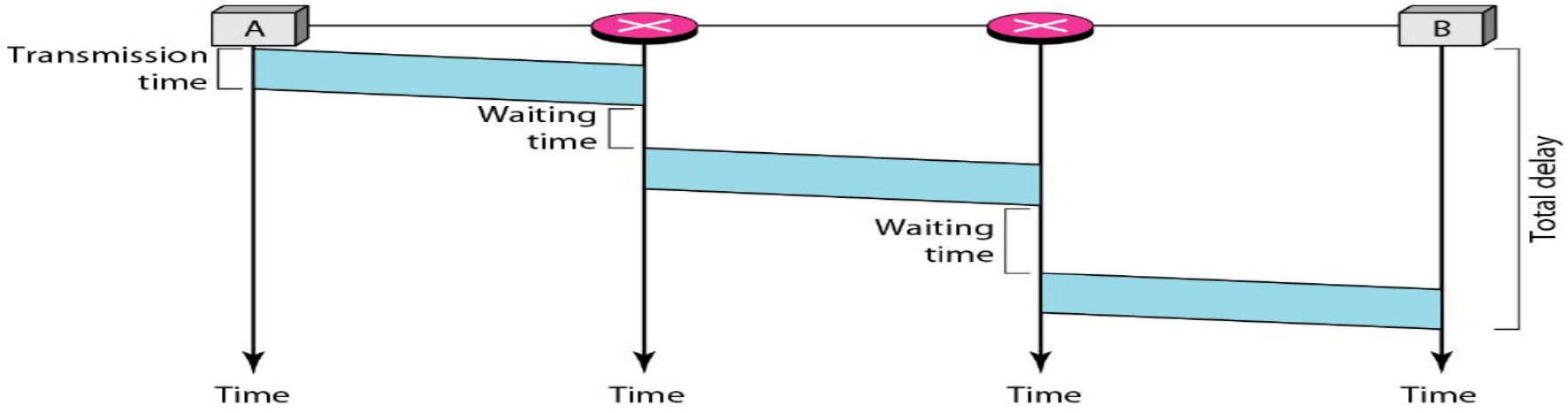


Efficiency

- Better than that of a circuit-switched network
- Resources are allocated only when there are packets to be transferred
- If a source sends a packet and there is a delay of a few minutes before another packet can be sent, the resources can be reallocated during these minutes for other packets from other sources.

Delay

- There are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded.
- Since not all packets in a message necessarily travel through the same switches, the delay is not uniform for the packets of a message.
- If the packet travels through two switches, there are three transmission times (3T), three propagation delays (slopes 3τ of the lines), and two waiting times (w1 + w2).
- We ignore the processing time in each switch. The total delay is
- **Total delay = 3T + 3τ + w1 + w2**



Virtual-Circuit Networks

- It is a cross between a circuit-switched network and a datagram network.
- As in a circuit-switched network, there are setup and teardown phases.
- Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
- As in a datagram network, data are packetized and each packet carries an address in the header.
- All packets follow the same path established during the connection.
- A virtual-circuit network is normally implemented in the data link layer
  - circuit-switched network is implemented in the physical layer
  - and a datagram network in the network layer.

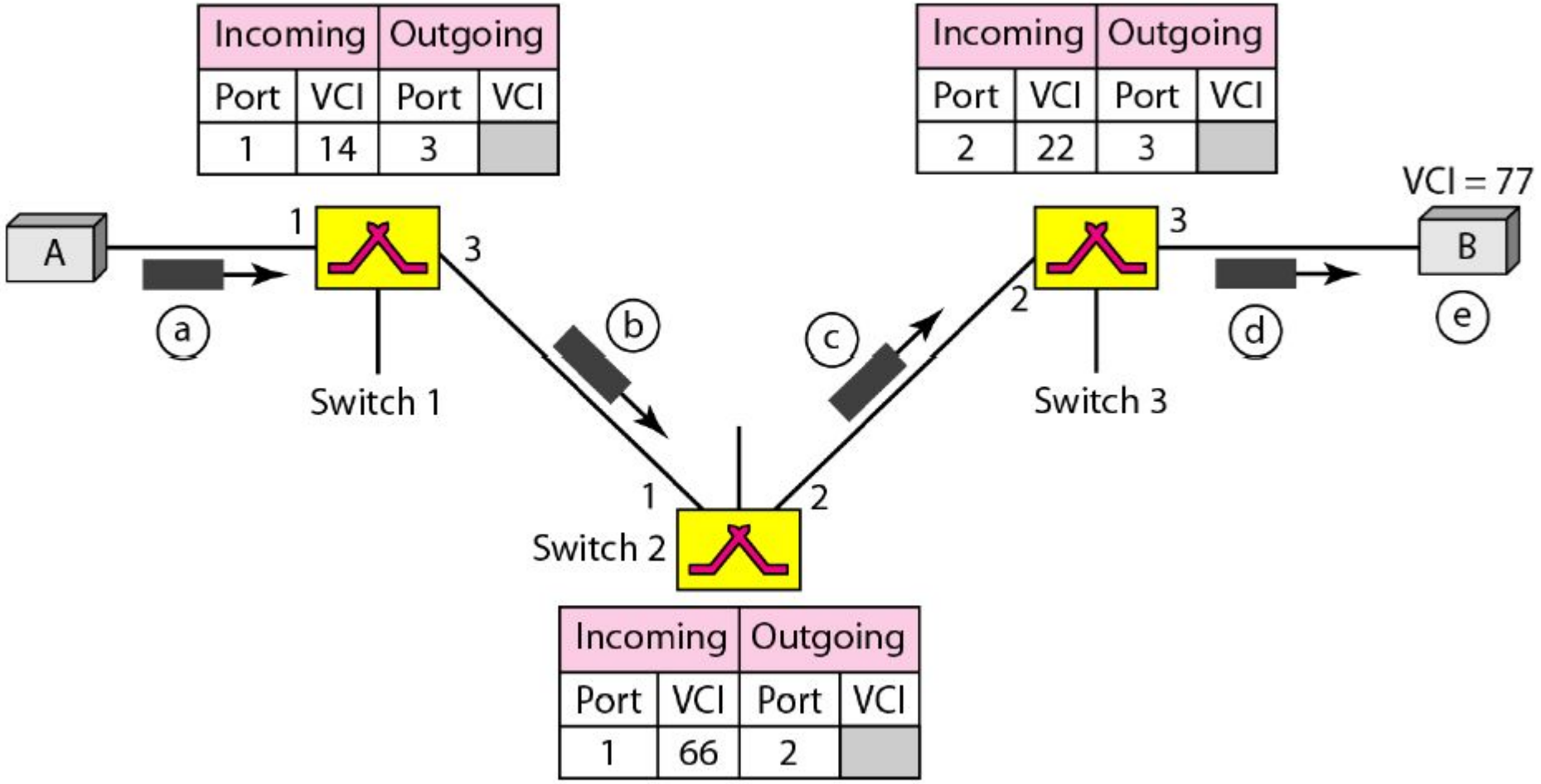
Addressing of VC

- Global Addressing
  - A source or a destination needs to have a global address that can be unique and is used only to create a virtual-circuit identifier
- Virtual-Circuit Identifier
  - The identifier that is actually used for data transfer is called the VCI
  - VCI does not need to be a large number since each switch can use its own unique set of VCIs

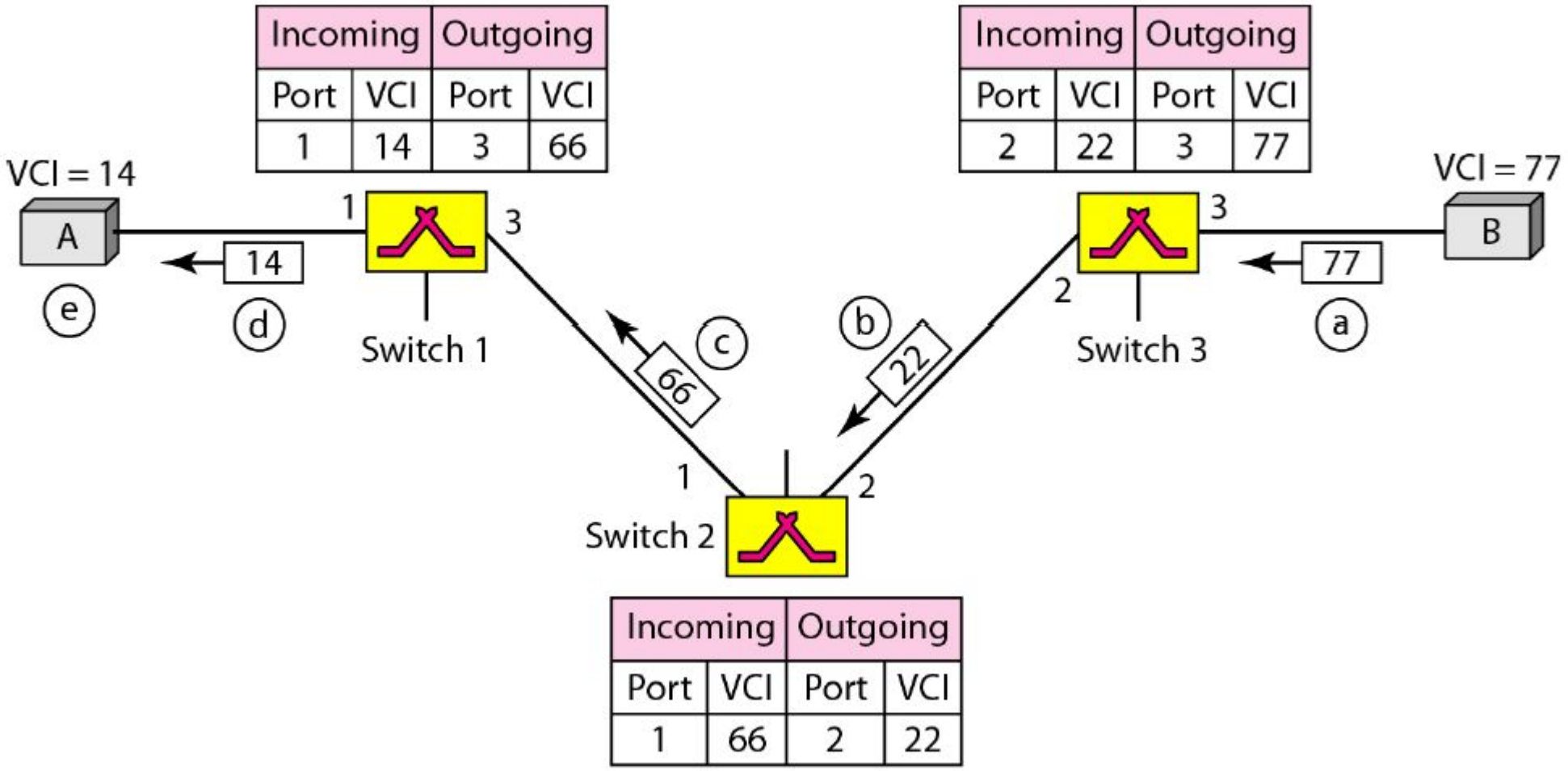
Three Phases of VC

- Setup phase:
  - the source and destination use their global addresses to help switches make table entries for the connection
- Data transfer phase:
  - occurs between these two phases.
- Teardown phase:
  - the source and destination inform the switches to delete the corresponding entry

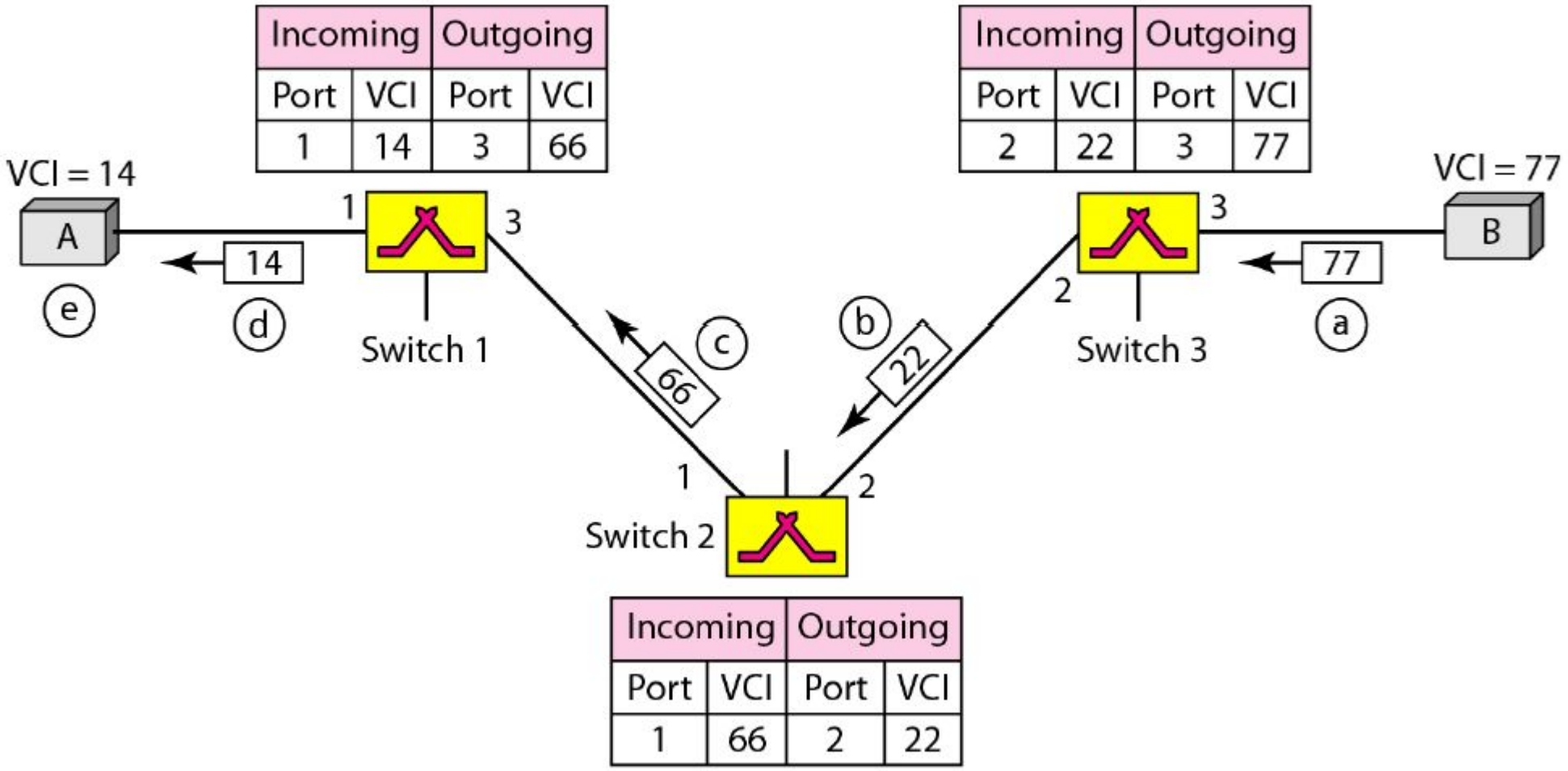
Setup phase (Setup Request)



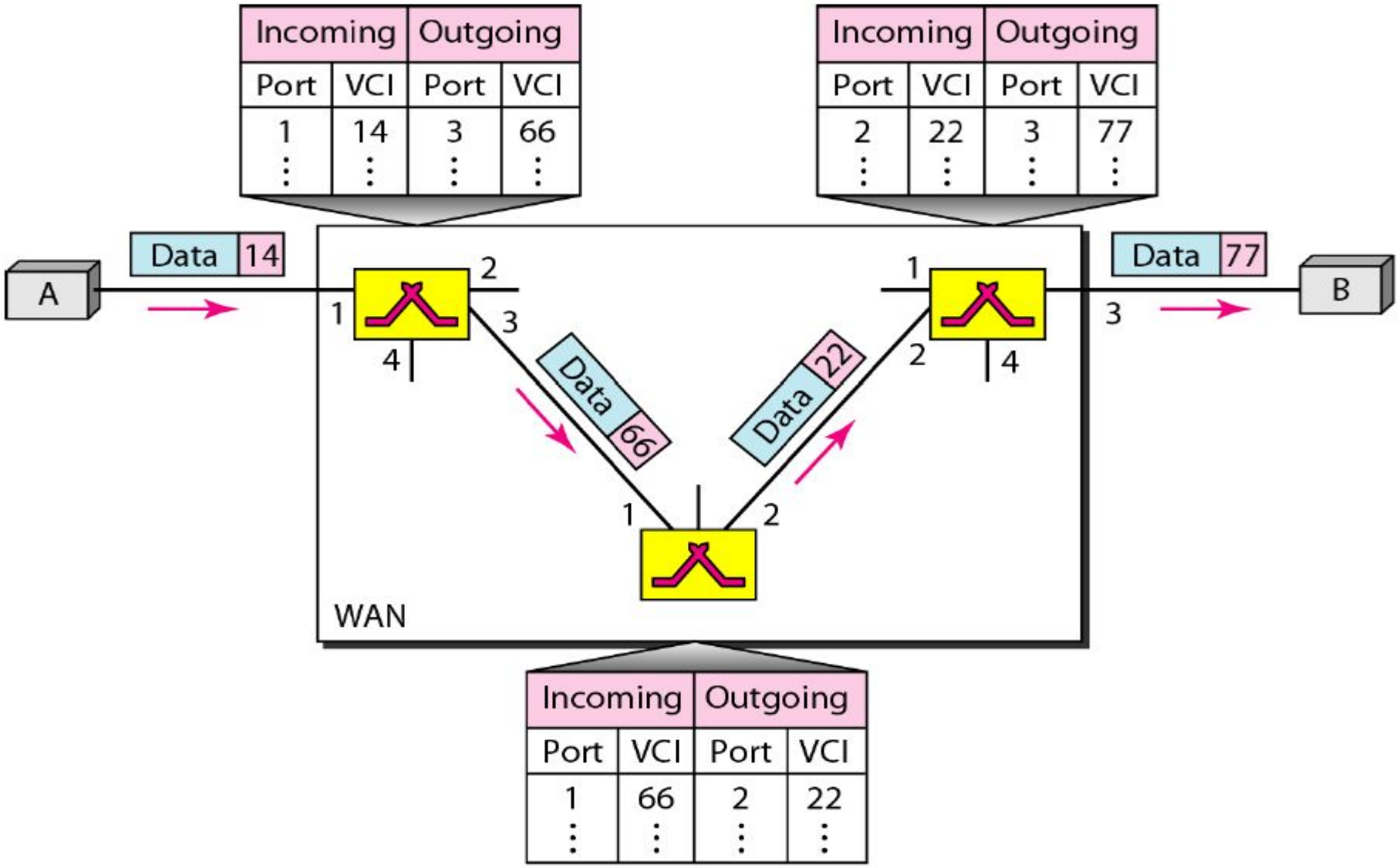
Setup phase (Acknowledgement)



Setup phase (Acknowledgement)



Data transfer Phase



Teardown Phase

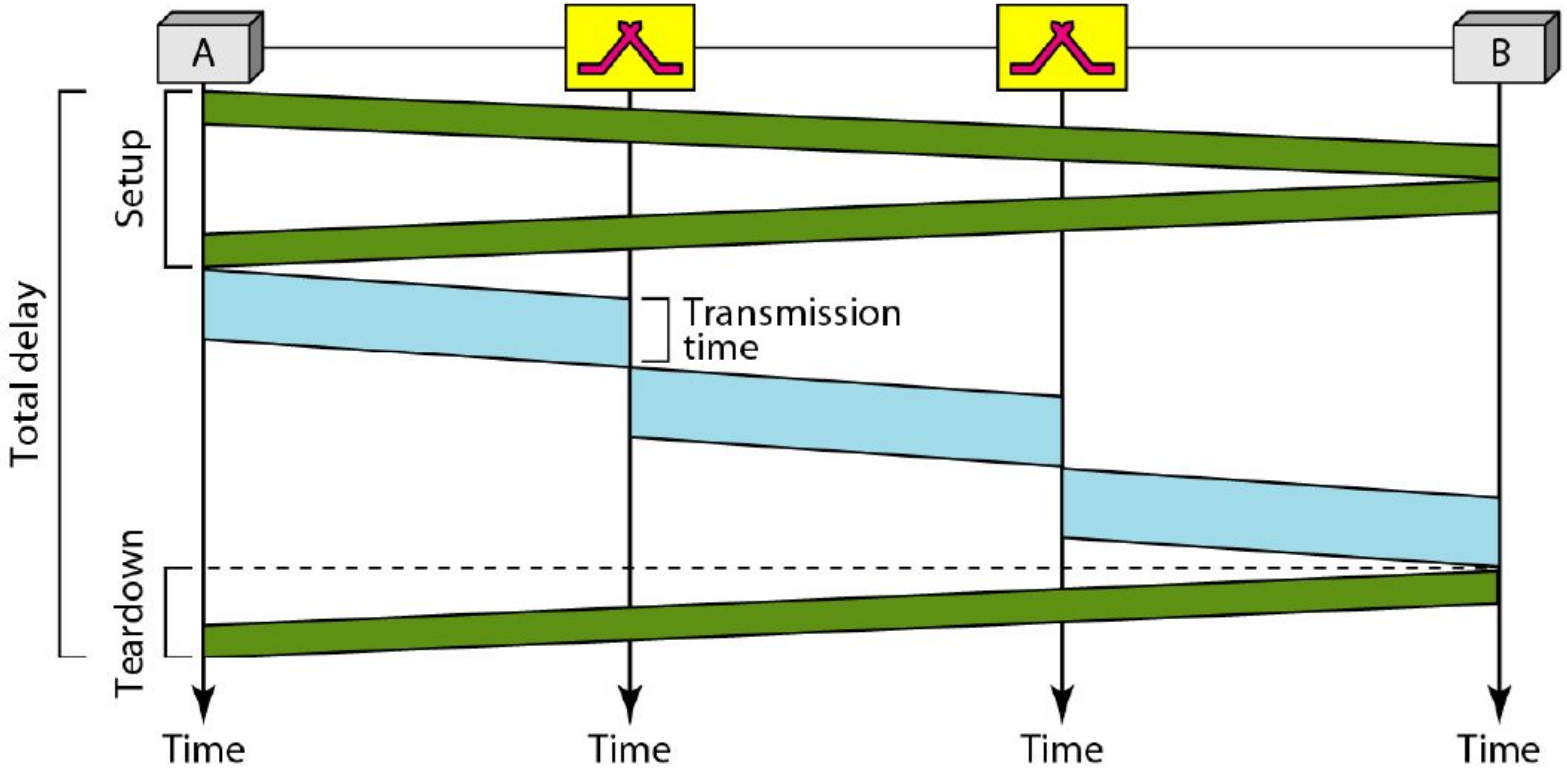
- In this phase, source A, after sending all frames to B, sends a special frame called a *teardown request*.
- Destination B responds with a teardown confirmation frame.
- All switches delete the corresponding entry from their tables.

Efficiency of VC

- Resource reservation in a virtual-circuit network can be made during the setup or on-demand during the data transfer phase
  - in first case, the delay for each packet is the same
  - in second case, each packet may encounter different delays
- Big advantage: even if resource allocation is on demand, The source can check the availability of the resources.

Delay in VC

- One-time delay for setup (in two direction)
- One-time delay for teardown (in one direction)
- If resources are allocated during setup phase, there is no wait time for individual packets.



The packet is traveling through two switches (routers). There are three transmission times (3T), three propagation times (3τ), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions),

Total delay =

3T+ 3τ + setup delay + teardown delay

ignore the processing time in each switch